

THE COGNITIVE BICYCLE

*Honest AI Use, Intellectual Ownership,
and the Framework the Academy Needs*

Matthew John Keen

Version 2.3

April 2026

Working Manuscript — Not Peer Reviewed

*AI, used correctly, is a compiler for original human thought. The human is the author. The tool
is the translator. Lineage is the record.*

Abstract

Generative artificial intelligence has altered the mechanics of scholarly production without eliminating human responsibility for intellectual direction. This manuscript argues that scholarly integrity depends not on rejecting new tools, but on documenting their use transparently. Scholarly trust rests on lineage — the traceable continuity between human intention, methodological development, and final output.

Drawing on historical precedent, emerging disclosure research, and structured evaluative reasoning, this work proposes lineage reporting as a constructive institutional response to AI-assisted intellectual work. It introduces the Compiler Analogy — framing AI as a translation tool for original human thought rather than a replacement for it — and demonstrates how such reasoning may be applied in practice.

Definition of Lineage

In this manuscript, lineage refers to the documented continuity between human intention, methodological development, and produced outcome across time.

Lineage does not describe tools. It describes responsibility.

It records how thought moves from origin to expression, regardless of the mechanisms used to articulate it.

Scope and Boundaries of This Framework

This manuscript does not attempt to:

- Detect AI-generated text
- Mechanically determine authorship
- Regulate or prohibit tool use
- Replace human scholarly judgement

Instead, it proposes a structured method for documenting intellectual lineage in AI-assisted scholarly workflows. The aim is clarification — not enforcement. Three boundaries follow from this directly:

1. This framework does not eliminate the need for human judgement.
2. It does not assign ownership automatically.
3. It does not assume that tool use reduces intellectual legitimacy.

Its purpose is documentation, not adjudication.

Part I — The Relocation of Skill

1.1 What the Luddites Misjudged

The Luddites were not irrational opponents of technology. They were skilled craftsmen responding to structural disruption. The power loom threatened the economic system within which their expertise had value. Their resistance was rational within that framework.

The lasting lesson is not resistance, but mislocation of value. They assumed value resided primarily in manual execution. In reality, the enduring skill lay in pattern recognition, material judgement, and production decisions.

Technological change does not eliminate skill. It relocates it.

1.2 The Same Structural Question in Academia

Generative AI systems produce fluent prose at speed. Academic evaluation has historically relied on written production as visible labour. This creates tension when production becomes automated.

If intellectual value resides in identifying problems, structuring reasoning, and defending conclusions, transcription alone cannot remain the primary signal of contribution.

The evaluative question is shifting:

Not: Who produced the text?
But: Who directed the thinking?

1.3 The Compiler Analogy — AI as Translation Tool

No one questions whether a programmer wrote code because they used a compiler. The compiler translates high-level human intent into machine code. The programmer provides the logic, architecture, and decisions. The compiler handles syntax, memory, and registers.

AI, used correctly, is a compiler for original human thought.

Traditional Compiler	AI — Used Correctly
Translates high-level code to machine code	Translates high-level thought to prose and structure
Human provides logic and architecture	Human provides ideas, judgements, and direction
Compiler handles syntax, memory, registers	AI handles grammar, phrasing, and drafting
Output is faithful to human intent	Output is faithful to human intent (when directed)

If a programmer is still the author of software they compiled, a writer using AI transparently is still the author of work they directed. The tool does not originate intent. The human does.

The analogy has a limit, and that limit matters. It holds only when the human provides the intent, structure, and judgement — when the AI is genuinely translating human thought rather than supplying it. Where that condition is not met, the analogy breaks down. This boundary is addressed directly in Part V.

1.4 Current Policy Measures

Major journals increasingly allow limited AI use under disclosure requirements. These policies regulate production behaviour but often fail to document reasoning lineage.

Disclosure of tool presence satisfies compliance. It does not constitute lineage documentation. A statement that AI was used tells a reader nothing about who originated the ideas, who structured the argument, or whether the author can defend the conclusions independently.

This distinction defines the problem addressed in this manuscript.

Part II — The Crisis of Disclosure

2.1 The Empirical Gap

The scale of AI involvement in scholarly writing is now measurable. Liang et al. (2025) estimate that approximately 13.5% of PubMed-indexed papers published in 2024 showed signs of LLM involvement — around 200,000 articles in a single year — with rates exceeding 20% in computer science preprints by late 2024.¹ Similar patterns appear across specialised domains, indicating structural rather than purely individual adoption.

Against this, disclosure rates remain strikingly low. Full-text analysis of 164,000 scientific publications found explicit AI disclosure in approximately 0.1% of papers published since 2023 — roughly 76 out of 75,000 papers.² The gap between use and declaration is not marginal. It is near-total.

This gap reflects adaptation lag, not necessarily ethical failure. Norms have not yet stabilised. Institutions have not yet provided sufficient guidance. Researchers are navigating ambiguity with incomplete tools.

¹ Liang, W., Zhang, Y., Wu, Z., et al. (2025). Quantifying Large Language Model Usage in Scientific Papers. *Nature Human Behaviour*, 9(12), 2599–2609.

² Academic journals' AI policies fail to curb the surge in AI-assisted academic writing. *Proceedings of the National Academy of Sciences* (2026). DOI: 10.1073/pnas.2526734123.

2.2 Structural Incentives and Incomplete Lineage

Binary reporting systems compress complex workflows into simplified declarations. Where assistance exists in partial form — as a sounding board, a drafting aid, a stress-test mechanism — ambiguity in the reporting requirement encourages silence.

Incomplete reporting weakens lineage continuity. Over time, this produces records that appear linear but conceal developmental complexity. The problem is not individual dishonesty but systemic inadequacy of the disclosure framework itself.

2.3 The Fabrication Principle

Engineering systems evolve through iteration:

design → failure → correction → documentation

When intermediate failures are omitted, final structures appear flawless from inception. Function remains. Lineage weakens. The structure is real; the history of how it came to be is not.

Scholarship faces a similar risk. When development stages disappear from records — the false starts, the reconceptualisations, the moments where a tool suggested something that was then evaluated, accepted, modified, or rejected — intellectual history becomes artificially simplified. Transparent lineage preserves the continuity that gives scholarship its credibility.

Part III — The Cognitive Bicycle

3.1 Race or Journey?

If a traveller must move from one place to another and a bicycle is available, using it is not deception. The bicycle extends reach.

Unless the traveller is in a race.

A race measures unaided effort. A journey measures arrival. The distinction is not about the bicycle. It is about what the activity is for.

Scholarship resembles a journey. Its measure is understanding — the quality of the insight reached, the rigour of the reasoning, the defensibility of the conclusions. It is not a measure of endurance of transcription.

3.2 The Bicycle as Cognitive Infrastructure

A large language model functions as conceptual infrastructure. It enables movement across domains that might otherwise require prolonged isolated study. It compresses the distance between a question and a useful response in the same way a bicycle compresses the distance between two points on a road.

Tool presence does not determine authorship. Direction determines authorship. The scholar who uses a language model to explore the literature on a topic, stress-test an argument, or articulate a thought more precisely is not less the author of the resulting work. The scholar who delegates the thinking to the tool entirely is not the author at all.

3.3 The Scholar as Rider

The rider provides what the bicycle cannot:

- Destination — the insight pursued
- Route — the sequence of reasoning
- Steering — judgement and correction
- Map — integrative understanding of where the work sits in the field

The bicycle expands movement. The rider determines meaning.

3.4 The Compiler and the Bicycle — A Unified Frame

The bicycle and the compiler appear to be different analogies. They are the same claim from two angles.

The bicycle addresses the question of effort: does using a tool constitute deception? The compiler addresses the question of authorship: does using a tool displace the human as the originating agent? Both answer no — under the same condition. The condition is direction. The rider provides destination; the programmer provides logic. Where that direction is present, the tool is a tool. Where it is absent, the tool has become the author.

The scholar is the rider. The AI is the bicycle. The compiler is the mechanism by which the bicycle moves. The human is the author.

3.5 Access and Intellectual Infrastructure

Institutional researchers benefit from libraries, supervision, peer networks, and structured seminar culture. Independent researchers may lack equivalent support — no supervisor to stress-test an argument, no peer group to challenge a framing, no library system providing deep literature access.

Used transparently, generative systems may partially substitute for structural access. They enable exploration, stress-testing, and cross-disciplinary connection that would otherwise require institutional affiliation. This does not eliminate access inequality, but it shifts its character.

Transparency preserves fairness. If an independent researcher uses AI to compensate for structural disadvantage and declares that use, the reader can evaluate the work on its merits. If the use is concealed, the reader cannot. Disclosure is therefore not merely an ethical obligation but a fairness mechanism.

Part IV — A Framework for Intellectual Ownership

4.1 Four Tests of Intellectual Ownership

The following tests formalise reasoning long used informally in scholarly defence. They are not new criteria; they are explicit statements of what examiners, editors, and peers have always implicitly assessed.

Test	Question	What it establishes
Spark Test	<i>Where did the initiating idea originate?</i>	Evidence must identify the human question or observation that began the work.
Map Test	<i>Who structured the argument?</i>	Ownership requires architectural direction — not merely generated structure.
Voice Test	<i>Whose judgement governs reasoning?</i>	Perspective, boundaries, and values must be identifiable and defensible.
Defense Test	<i>Can the work be defended independently of the tool?</i>	Understanding remains the final test of intellectual responsibility.

These tests do not automate judgement. They structure it. They provide a common vocabulary for a conversation that is currently happening without shared terms.

They also define the boundary of the compiler analogy. The analogy holds when a scholar can pass all four tests. It breaks down when they cannot — when there is no identifiable Spark, no human Map, no defensible Voice, and no capacity for Defense. At that point the work has not been compiled; it has been delegated.

4.2 The Lineage Declaration

AI-assisted work should include structured provenance documentation recording:

- Concept origin — where the initiating question came from
- Stages of tool involvement — at what points and for what purposes the tool was used
- Boundaries of assistance — what the tool did and did not contribute
- Independent defence capability — confirmation that the author can explain and defend all claims

This converts invisible workflow into visible lineage. Documentation strengthens trust without restricting innovation. It does not penalise tool use. It penalises undisclosed tool use.

4.3 Integrity Questions for AI-Assisted Work

Three practical questions operationalise lineage reasoning in everyday scholarly practice:

1. Is the core idea traceable to identifiable human inquiry?
2. Is tool involvement declared with sufficient clarity for a reader to evaluate the work?
3. Can the reasoning be defended without reliance on the tool?

These questions are not a checklist. They are a test of whether the lineage declaration holds under scrutiny.

4.4 Evaluation Practice

As generated text becomes indistinguishable from human prose, detection tools lose reliability. The arms race between generation and detection produces no durable winner. Detection-based evaluation is therefore not a sustainable institutional response.

Structured challenge — oral examination, targeted questioning on methodology and reasoning, requests for explanation of specific claims — provides stronger evaluation. Final judgement remains human. The question is not whether the text looks human. The question is whether the author understands it.

4.5 Institutional Responsibility

Institutions stabilise knowledge systems through documentation of change. When transitions occur without record — when new tools are adopted without acknowledgement, when workflows shift without disclosure frameworks — retrospective coherence replaces historical accuracy.

Reliable knowledge emerges from visible correction. Trust depends on recorded development. Institutions that establish clear lineage reporting requirements do not restrict scholarship. They protect it.

Part V — Counterarguments and Limits

Analytical integrity requires acknowledgement of limitations. Four are considered here.

AI Novelty

Generative systems may occasionally produce unexpected conceptual arrangements — connections, formulations, or juxtapositions that the human author did not anticipate. This raises a genuine question about whether such outputs constitute a form of machine creativity.

Intellectual ownership, however, remains contingent on human recognition, evaluation, and defence of those arrangements. Unexpected output does not constitute authorship. The human who encounters an unexpected arrangement and judges it valuable, incorporates it, develops it, and can defend it has exercised the authorial function. The system that produced it has not.

Writing Cognition

Manual writing supports reflective cognition in ways that are not fully understood. The physical act of composition — selecting each word, constructing each sentence — may itself generate thought rather than merely record it. Reducing transcription effort through AI assistance may therefore alter reasoning patterns in ways that are not captured by the finished output.

This remains an open empirical question. The framework distinguishes between articulation and understanding, but does not assume they are fully separable. Scholars who find that the act of writing is itself a mode of thinking should treat AI assistance in drafting accordingly — as a tool that may accelerate production but may also short-circuit a productive cognitive process.

Access Inequality

Advanced tools are not uniformly accessible. Paywalled services, hardware requirements, and language bias in training data all introduce new disparities. The claim that AI democratises access to intellectual infrastructure must be qualified by acknowledging that access to AI is itself unequal.

Transparent declaration does not eliminate inequality. It exposes it. That exposure creates the conditions for institutional adaptation — for identifying where access gaps exist and what structural responses they require.

The Compiler Limit

Not all AI use is compiler-like. This is the most important limitation of the central analogy, and it must be stated plainly.

Using AI to generate core arguments without direction, review, or defence is not compilation — it is delegation. The compiler analogy holds only when the human provides the intent, structure, and judgement, and when the human can pass all four ownership tests: Spark, Map, Voice, and Defense. Without those, the human is not the author. The analogy does not rescue undirected delegation by relabelling it as compilation.

The framework proposed in this manuscript is therefore not a defence of all AI use. It is a defence of directed, transparent, documented AI use — and a demand for honesty about where that threshold has not been met.

Part VI — Illustrative Application to This Work

The framework is applied here to its own production as demonstration rather than validation. Self-application is not proof of validity. It is an illustration of what lineage documentation looks like in practice.

6.1 On the Origin of This Work — ARKive and the Question of Lasting Record

The conceptual origins of this manuscript lie in two encounters with the same problem from different directions.

The first was a visit to Microsoft, where work on long-duration archival storage technology was under discussion. The question of how digital material might be preserved across centuries — rather than merely decades — was being addressed at a technical level. That encounter prompted further investigation into what such technology might mean for fields where the survival of records was not merely a technical problem but an intellectual one.

The second was involvement with ARKive. ARKive was a Millennium-funded global initiative — supported by the UK Heritage Lottery Fund and the New Opportunities Fund as part of the UK's Millennium celebrations — with the mission of creating a centralised digital archive of the world's endangered species, drawing on the work of over 7,000 wildlife filmmakers, photographers, conservationists, and scientists. At its height it served over half a million users monthly from more than 200 countries.

In February 2019, ARKive went offline. The decision was made by Wildscreen, the UK charity that ran the project, due to sustained funding failure — despite a user base of 5.6 million, only 85 users made donations in 2018. The content — over 100,000 images and videos — was preserved offline. The public access was not.

ARKive is a precise illustration of the distinction this manuscript draws between tool presence and lineage preservation. The material survived. The continuity of access did not. The knowledge existed; the record of how to reach it was severed. The Microsoft encounter had raised the

technical question of how long a record can last. ARKive demonstrated that the limiting factor is rarely the medium. It is the institutional will to maintain the chain.

This manuscript is, in part, an attempt to think clearly about what it means to preserve intellectual lineage in conditions where both the tools and the institutions that sustain them are unreliable. That question did not arise from a literature review. It arose from watching something fail that should not have.

6.2 Ownership Tests Applied

Test	Application to This Work
Spark	Human-origin conceptual inquiry rooted in encounters with the problem of digital preservation across two distinct contexts: a visit to Microsoft where long-duration archival storage technology was under development, and subsequent involvement with ARKive — a Millennium-funded global wildlife archive that went offline in 2019 despite its content surviving. The initiating question arose from witnessing how lineage can be preserved in the medium while access is lost in the institution. See Section 6.1.
Map	Human-directed architecture determining conceptual sequence, argument flow, and integration of the compiler analogy across all seven parts.
Voice	Ethical commitment to transparency derived from lived experience as an independent researcher operating without institutional support structures.
Defense	Independent explanatory capacity across all framework components, including the compiler analogy, the bicycle frame, and the four lineage tests.

6.3 Declared Tool Use

A large language model was used in the production of this work in the following capacities:

- Research assistant — exploring relevant literature and surfacing connections across fields
- Critical interlocutor — stress-testing arguments, identifying weaknesses, and proposing counterarguments
- Drafting instrument — translating developed ideas into structured prose
- Cross-disciplinary connector — identifying analogues in engineering, history of technology, and archival science

The tool did not originate the ideas. It did not determine the structure. It cannot defend the argument.

The AI was the compiler. The human was the author.

Part VII — Conclusion

Generative tools alter scholarly mechanics but not intellectual responsibility. The central challenge posed by AI-assisted knowledge production is not technological control. It is lineage preservation — ensuring that the chain from human intention to produced output remains visible, documented, and defensible.

Scholarship survives technological change when responsibility remains visible. The printing press did not dissolve authorship. The word processor did not dissolve it. The compiler did not dissolve it. Generative AI will not dissolve it either — provided that institutions insist on, and scholars practise, transparent documentation of intellectual lineage.

Tools expand capability, but lineage preserves truth. Where capability expands without documentation, confusion follows — not immediately, but over time, as the distance between what was claimed and what was independently understood widens without anyone noticing. Where capability expands with transparent lineage, scholarship remains reliable because the record of how knowledge was produced survives alongside the knowledge itself.

AI, used correctly, is a compiler for original human thought. The human is the author. The tool is the translator. Lineage is the record.

This is not a concession to technology. It is an affirmation of human direction, made visible through honest documentation.

ARKive went offline. The content survived. What was lost was not the material but the continuity of access — the chain that connected the work to the world. The lesson is not that digital

preservation fails. It is that lineage requires maintenance. That maintenance is institutional, financial, and human. No medium, however durable, preserves a record that no one has chosen to keep.

That choice — to keep the record, to document the chain, to make the lineage visible — is what this framework asks of scholars, and what it asks of the institutions that sustain them.

Appendix — Summary of Key Distinctions

Traditional concern	Lineage-based response
<i>"Did you use AI?"</i>	"Did you direct the thinking?"
<i>"Who produced the text?"</i>	"Who structured the argument?"
<i>"Can we detect AI?"</i>	"Can you defend the work?"
<i>"Is tool use cheating?"</i>	"Is lineage documented?"
<i>"AI writes instead of you"</i>	"AI compiles for you"